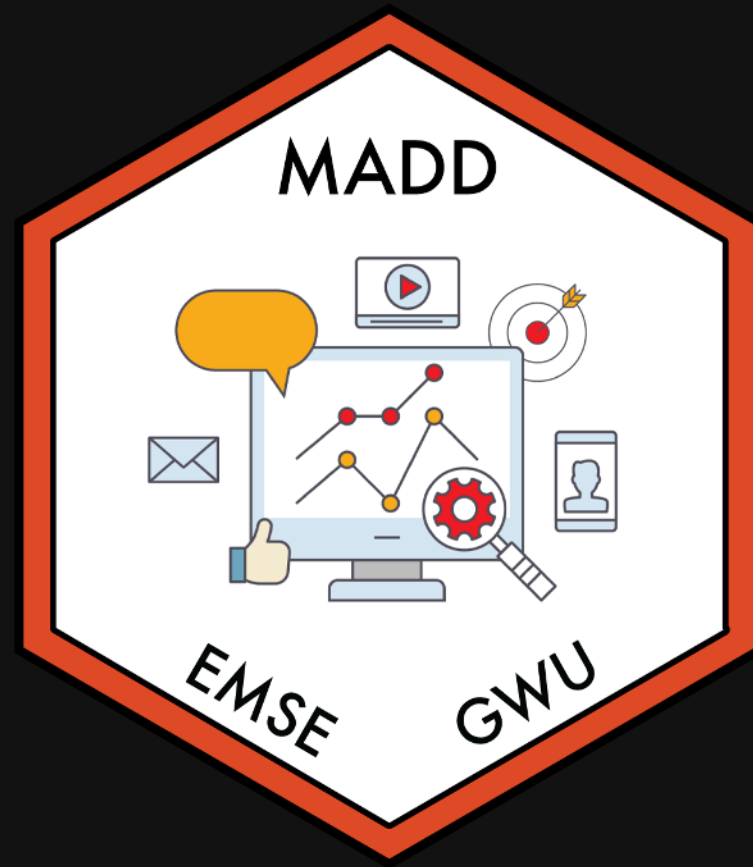




Quarto & Plotting

🏛️ EMSE 6035: Marketing Analytics for Design Decisions

👤 John Paul Helveston

📅 September 23, 2026

Week 5: *Quarto & Plotting*

1. Intro to Quarto
2. Intro to ggplot2
3. Project attributes & levels
4. Conjoint survey components

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“Literate programming”

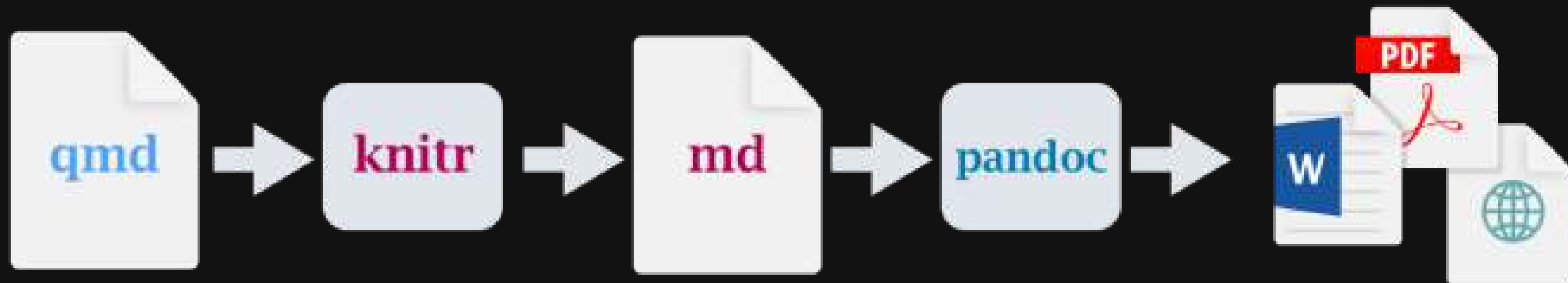
Treat programs as a “literature”
understandable to **human
beings**



Donald E. Knuth

Quick demo

1. Open `quarto_demo.qmd`
2. Click “Preview”



Anatomy of a .qmd file

Header

Markdown text

R code

Define overall document options in header

Basic html page

```
1 ----
2 title: Your title
3 author: Author name
4 format: html
5 ----
```

Add table of contents, change theme

```
1 ----
2 title: Your title
3 author: Author name
4 toc: true
5 format:
6   html:
7     theme: united
8 ----
```

More on themes at

<https://quarto.org/docs/output-formats/html-themes.html>

Render to multiple outputs

PDF uses LaTeX

```
1 ----  
2 title: Your title  
3 author: Author name  
4 format: pdf  
5 ----
```

If you don't have LaTeX on your computer, install tinytex in R:

```
1 tinytex::install_tinytex()
```

Microsoft Word

```
1 ----  
2 title: Your title  
3 author: Author name  
4 format: docx  
5 ----
```

Anatomy of a .qmd file

~~Header~~

Markdown text

R code

Right now, bookmark this! 🖱️

<https://commonmark.org/help/>

(When you have 10 minutes, do this! 🖱️)

<https://commonmark.org/help/tutorial/>

Headers

```
1 # HEADER 1
2
3 ## HEADER 2
4
5 ### HEADER 3
6
7 #### HEADER 4
8
9 ##### HEADER 5
10
11 ##### HEADER 6
```

HEADER 1

HEADER 2

HEADER 3

HEADER 4

HEADER 5

HEADER 6

Basic Text Formatting

Type this...

- `normal text`
- `_italic text_`
- `*italic text*`
- `**bold text**`
- `***bold italic text***`
- `~~strikethrough~~`
- ``code text``

..to get this

- normal text
- *italic text*
- *italic text*
- **bold text**
- ***bold italic text***
- ~~strikethrough~~
- `code text`

Lists

Bullet list:

```
1 - first item  
2 - second item  
3 - third item
```

- first item
- second item
- third item

Numbered list:

```
1 1. first item  
2 2. second item  
3 3. third item
```

1. first item
2. second item
3. third item

Links

Simple **url link** to another site:

```
1 [Download R](http://www.r-project.org/)
```

[Download R](http://www.r-project.org/)

Anatomy of a .qmd file

~~Header (think of this as the “settings”)~~

~~Markdown text~~

R code

R Code

Inline code

```
1 `r insert code here`
```

Code chunks

```
1 ```{r}  
2 insert code here  
3 insert more code here  
4 ```
```

Inline R code

```
1 The sum of 3 and 4 is `r 3 + 4`
```

Produces this:

The sum of 3 and 4 is 7

R Code chunks

This code chunk...

```
1 ```{r}
2 library(palmerpenguins)
3
4 glimpse(penguins)
5 ```
```

...will produce this when rendered:

```
1 library(palmerpenguins)
2
3 glimpse(penguins)
```

```
#> Rows: 4
#> Columns: 8
#> $ species      <fct> Adelie, Adelie, Adelie, Adelie
#> $ island       <fct> Torgersen, Torgersen, Torgersen, Tor
#> $ bill_length_mm <dbl> 39.1, 39.5, 40.3, NA
#> $ bill_depth_mm <dbl> 18.7, 17.4, 18.0, NA
#> $ flipper_length_mm <int> 181, 186, 195, NA
#> $ body_mass_g   <int> 3750, 3800, 3250, NA
#> $ sex           <fct> male, female, female, NA
#> $ year          <int> 2007, 2007, 2007, 2007
```

Chunk options

Control what chunks output using options

All options [here](#)

option	default	effect
eval	TRUE	Whether to evaluate the code and include its results
echo	TRUE	Whether to display code along with its results
warning	TRUE	Whether to display warnings
error	FALSE	Whether to display errors
message	TRUE	Whether to display messages
tidy	FALSE	Whether to reformat code in a tidy way when displaying it
results	"markup"	"markup", "asis", "hold", or "hide"
cache	FALSE	Whether to cache results for future renders
comment	"##"	Comment character to preface results with
fig.width	7	Width in inches for plots created in chunk
fig.height	7	Height in inches for plots created in chunk

Chunk output options

By default, code chunks print **code** + **output**

```
1 ```{r}
2 #| echo: false
3
4 cat('hello world!')
5 ```
```

Prints only **output**
(doesn't show code)

```
#> hello world!
```

```
1 ```{r}
2 #| eval: false
3
4 cat('hello world!')
5 ```
```

Prints only **code**
(doesn't run the code)

```
1 cat("hello world!")
```

```
1 ```{r}
2 #| include: false
3
4 cat('hello world!')
5 ```
```

Runs, but doesn't print
anything

A global `setup` chunk

```
1  ```{r}
2  #| label: setup
3  #| include: false
4
5  knitr::opts_chunk$set(
6    warning = FALSE,
7    message = FALSE,
8    fig.width = 7.252,
9    fig.height = 4,
10   comment = "#>"
11 )
12  ```
```

- Typically the first chunk
- All following chunks will use these options (i.e., sets global chunk options)
- You can (and should) use individual chunk options too
- Often where I load libraries, etc.

Your turn

15:00

1. Open the `bears.qmd` file, and title it *"Bears Analysis"*
2. Create a "setup" code chunk to read in the `bear_killings.csv` data file (HINT: You might want to look back at the `quarto_demo.qmd` file!).
3. Use text and code to find answers each of the following questions - show your code and results to justify each answer:
 - Which months have the highest frequency of bear killings?
 - Who has been killed more often by bears: hunters or hikers?
 - How do the the number of bear attacks on men vs women compare?

HINT: Use `bears %>% count(variable)` to count how many rows are in the data for each unique value of `variable`

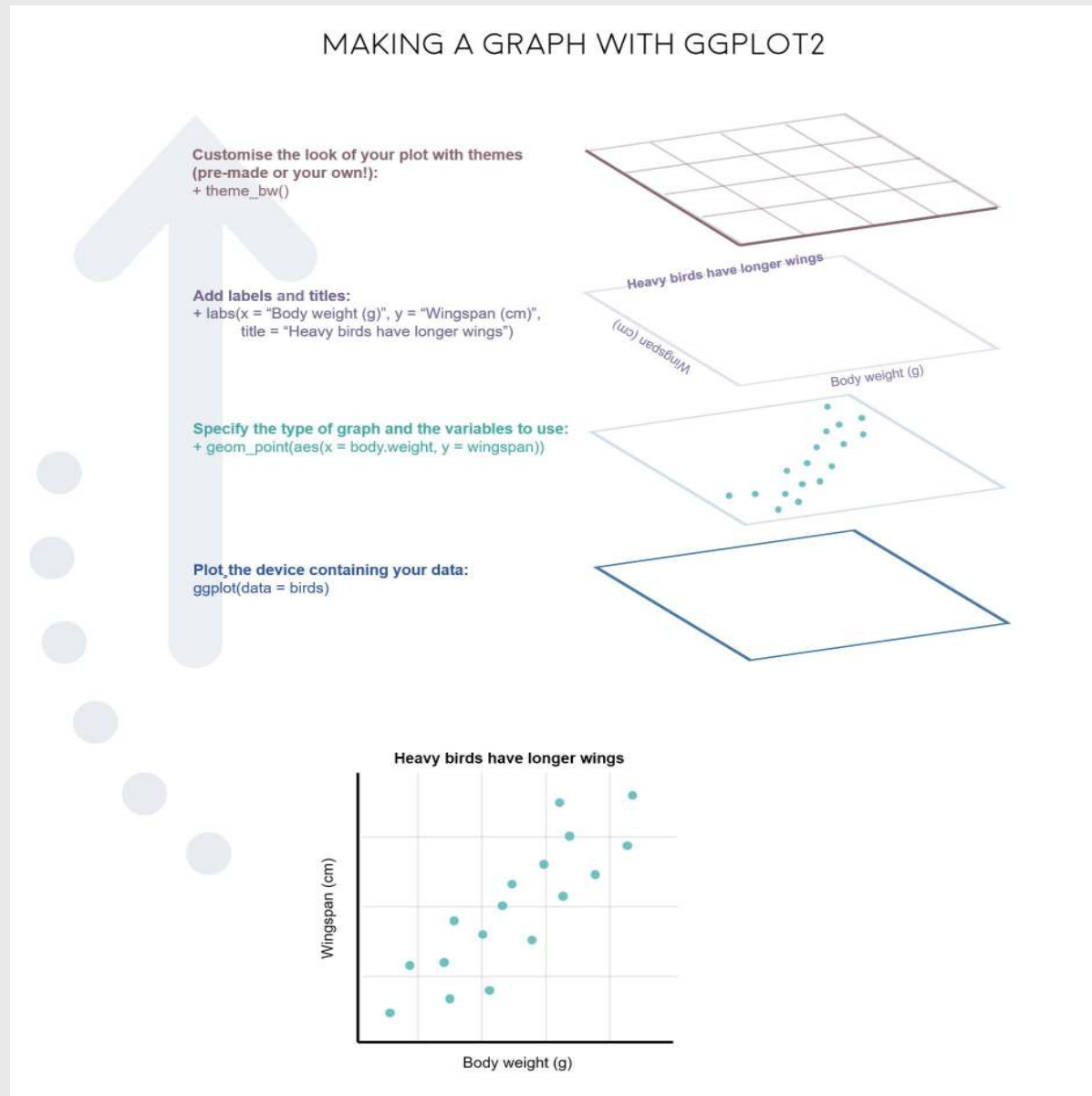
Week 5: *Quarto & Plotting*

1. Intro to Quarto
2. Intro to `ggplot2`
3. Project attributes & levels
4. Conjoint survey components

“Grammar of Graphics”

Concept developed by Leland Wilkinson (1999)

ggplot2 package developed by Hadley Wickham (2005)



Making plot layers with ggplot2

1. The data
2. The aesthetic mapping (what goes on the axes?)
3. The geometries (points? bars? etc.)
4. The annotations / labels
5. The theme

Layer 1: The data

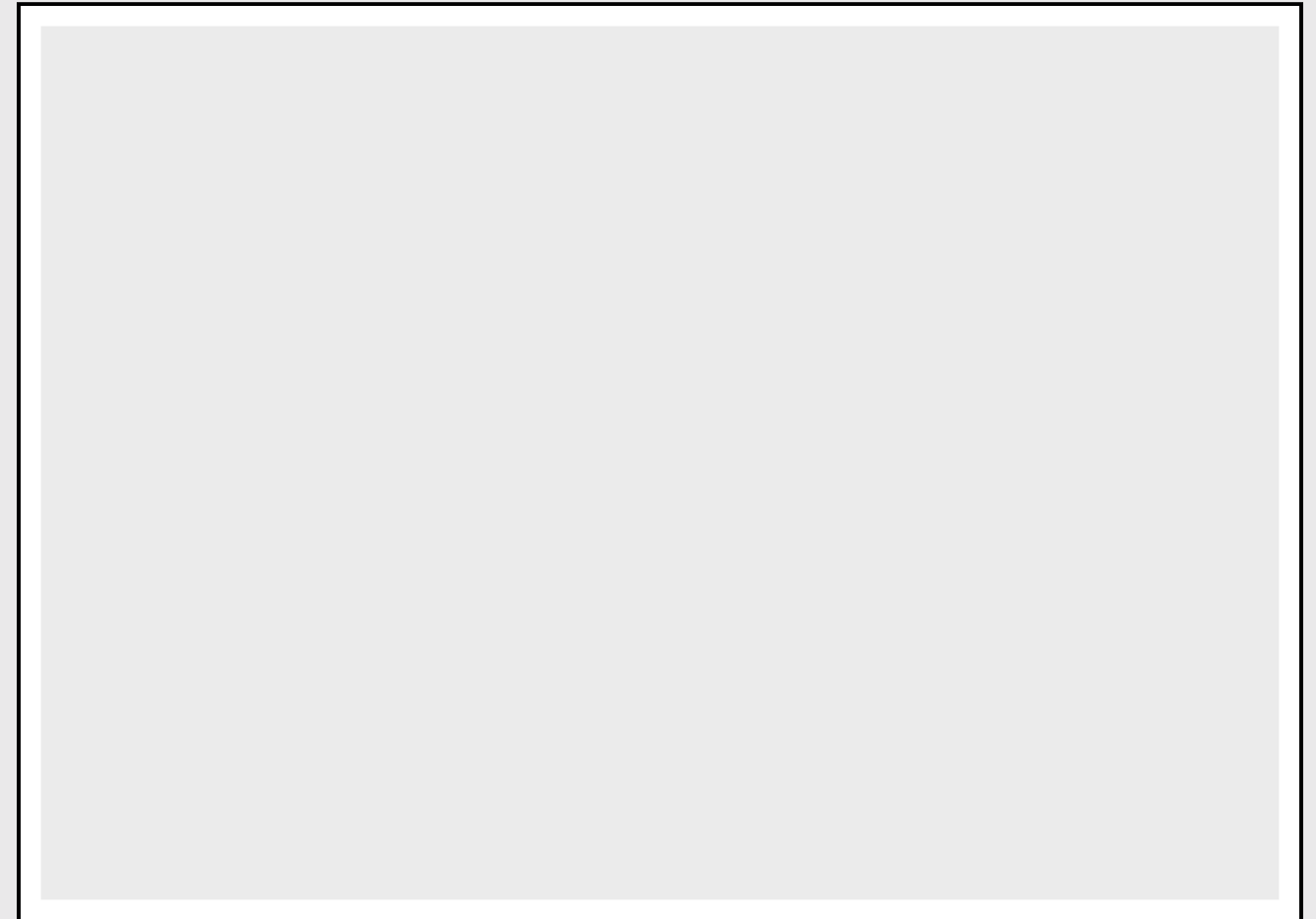
```
1 head(mpg)
```

```
#> # A tibble: 6 × 11
#>   manufacturer model displ  year   cyl trans      drv    cty   hwy fl      class
#>   <chr>          <chr> <dbl> <int> <int> <chr>    <chr> <int> <int> <chr> <chr>
#> 1 audi          a4      1.8  1999     4 auto(l5)  f       18    29 p      compact
#> 2 audi          a4      1.8  1999     4 manual(m5) f       21    29 p      compact
#> 3 audi          a4      2    2008     4 manual(m6) f       20    31 p      compact
#> 4 audi          a4      2    2008     4 auto(av)   f       21    30 p      compact
#> 5 audi          a4      2.8  1999     6 auto(l5)  f       16    26 p      compact
#> 6 audi          a4      2.8  1999     6 manual(m5) f       18    26 p      compact
```

Layer 1: The data

The `ggplot()` function initializes the plot with whatever data you're using

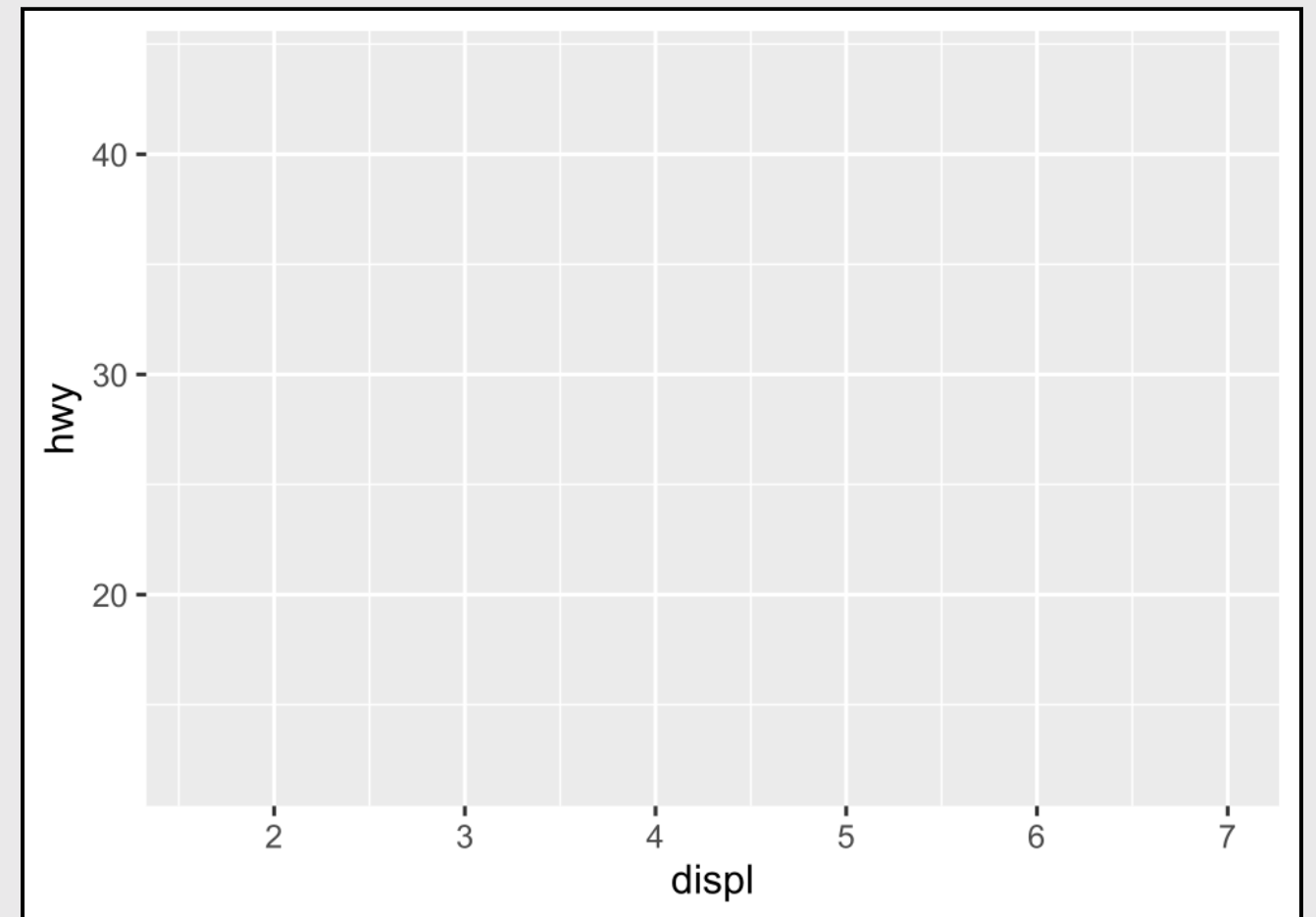
```
1 mpg %>%  
2   ggplot()
```



Layer 2: The aesthetic mapping

The `aes()` function determines which variables will be *mapped* to the geometries (e.g. the axes)

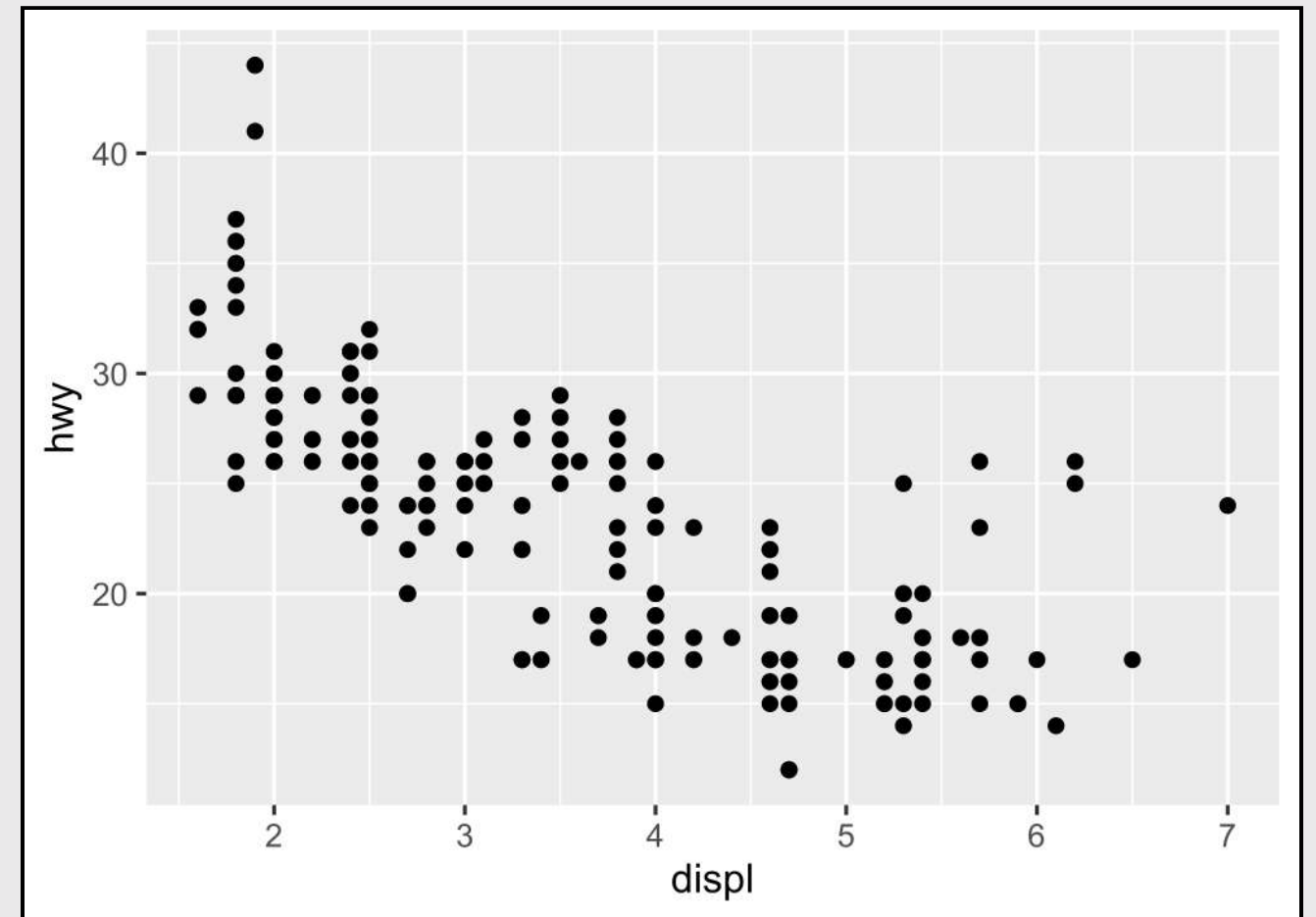
```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy))
```



Layer 3: The geometries

Use `+` to add geometries, e.g. `geom_points()` for points

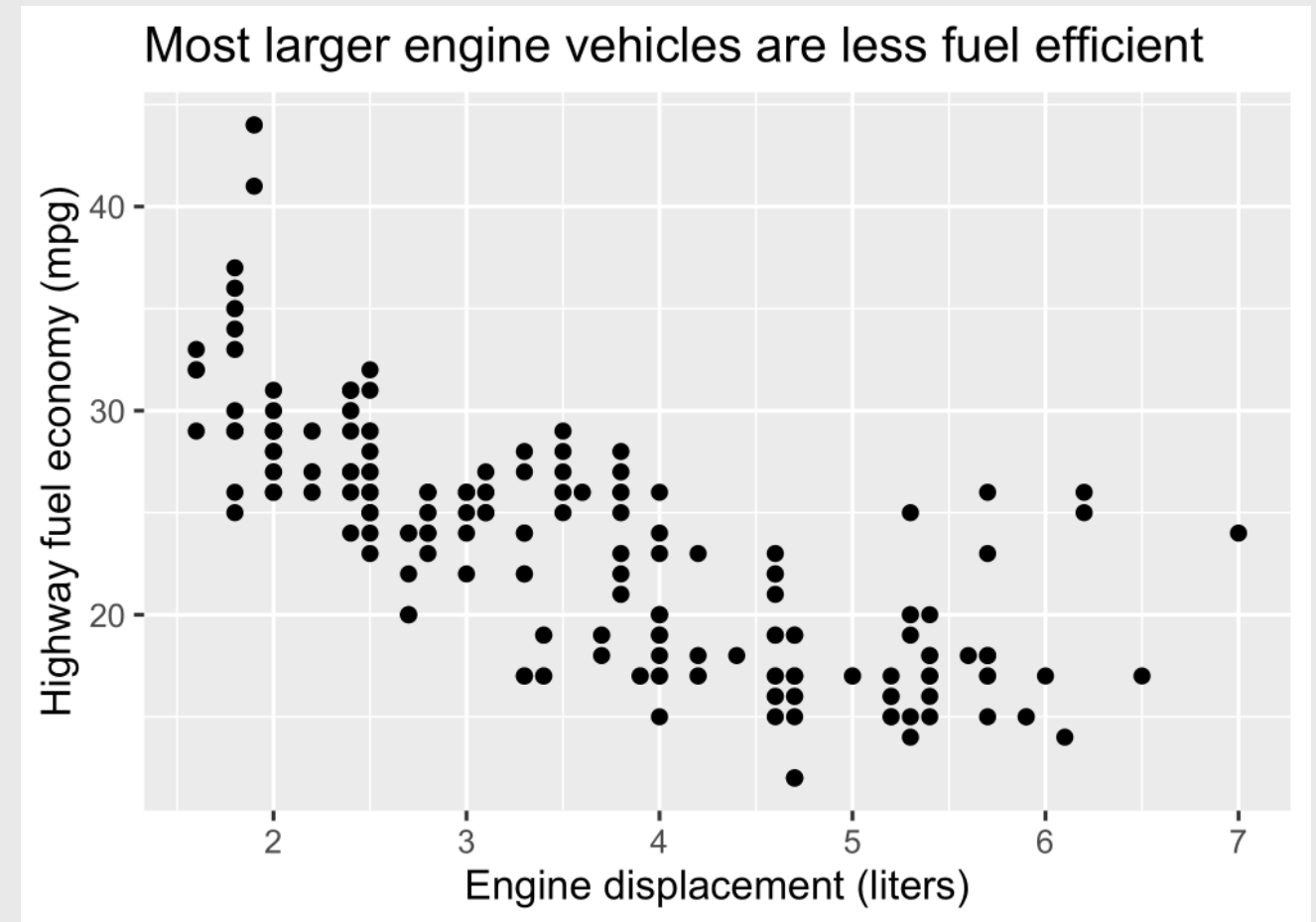
```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point()
```



Layer 4: The annotations / labels

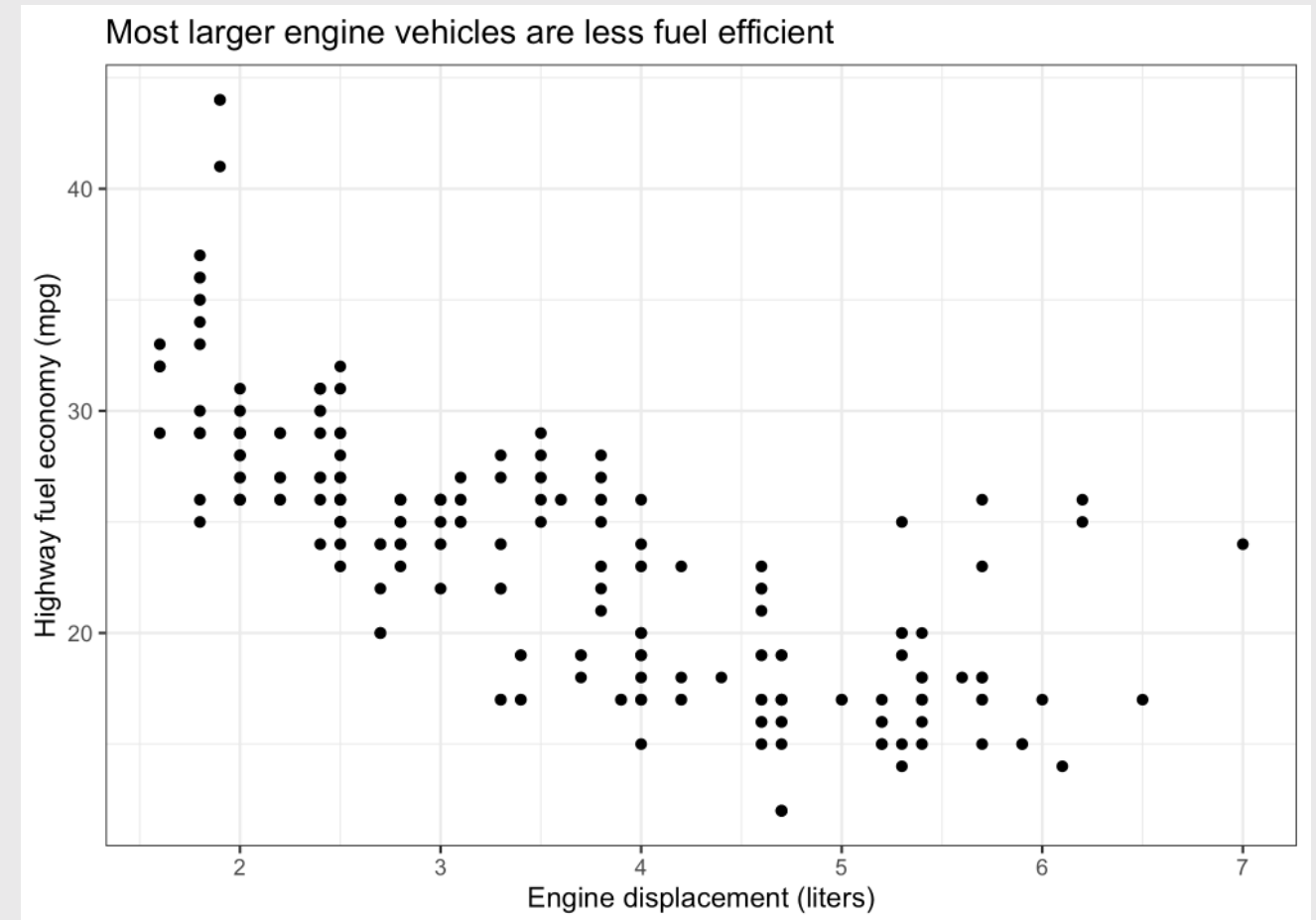
Use `labs()` to modify most labels

```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point() +  
4   labs(  
5     x = "Engine displacement (liters)",  
6     y = "Highway fuel economy (mpg)",  
7     title = "Most larger engine vehicles are  
8   )
```



Layer 5: The theme

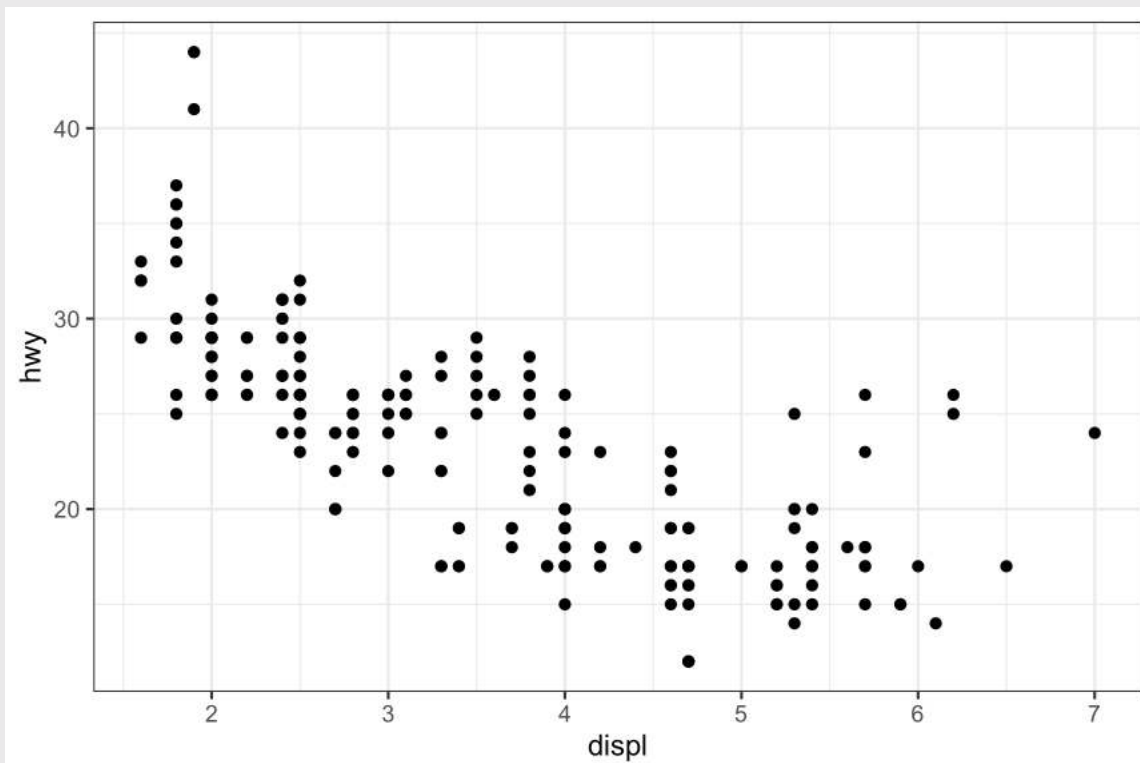
```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point() +  
4   labs(  
5     x = "Engine displacement (liters)",  
6     y = "Highway fuel economy (mpg)",  
7     title = "Most larger engine vehicles are  
8   ) +  
9   theme_bw()
```



Common themes

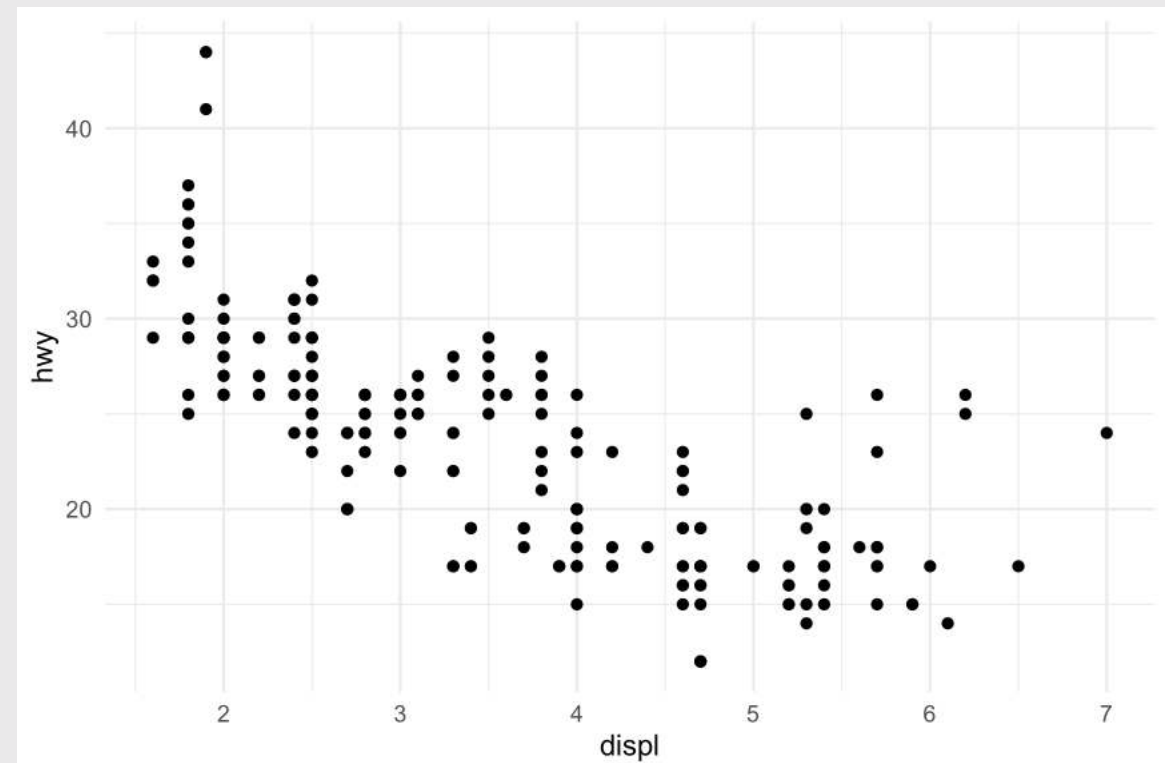
theme_bw()

```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point() +  
4   theme_bw()
```



theme_minimal()

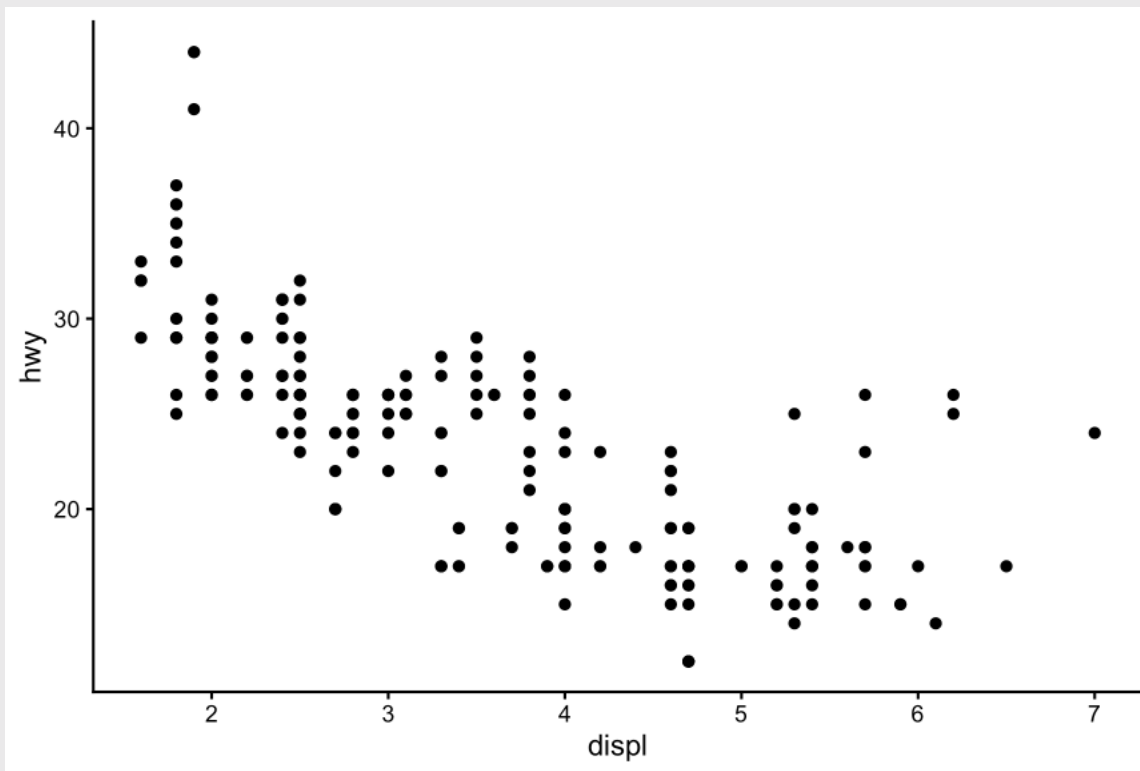
```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point() +  
4   theme_minimal()
```



Common themes

theme_classic()

```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point() +  
4   theme_classic()
```



theme_void()

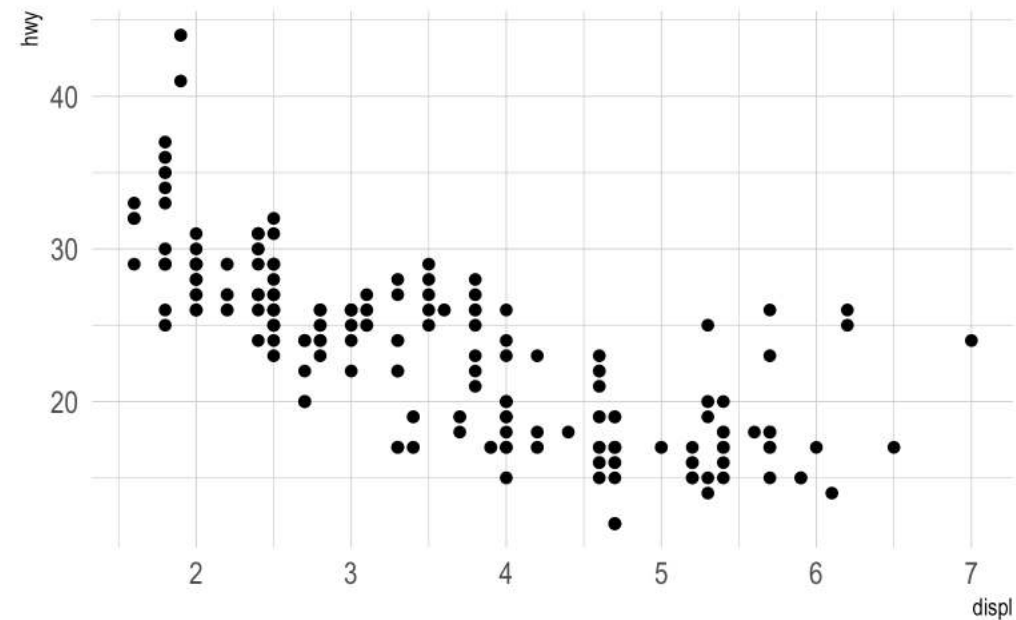
```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point() +  
4   theme_void()
```



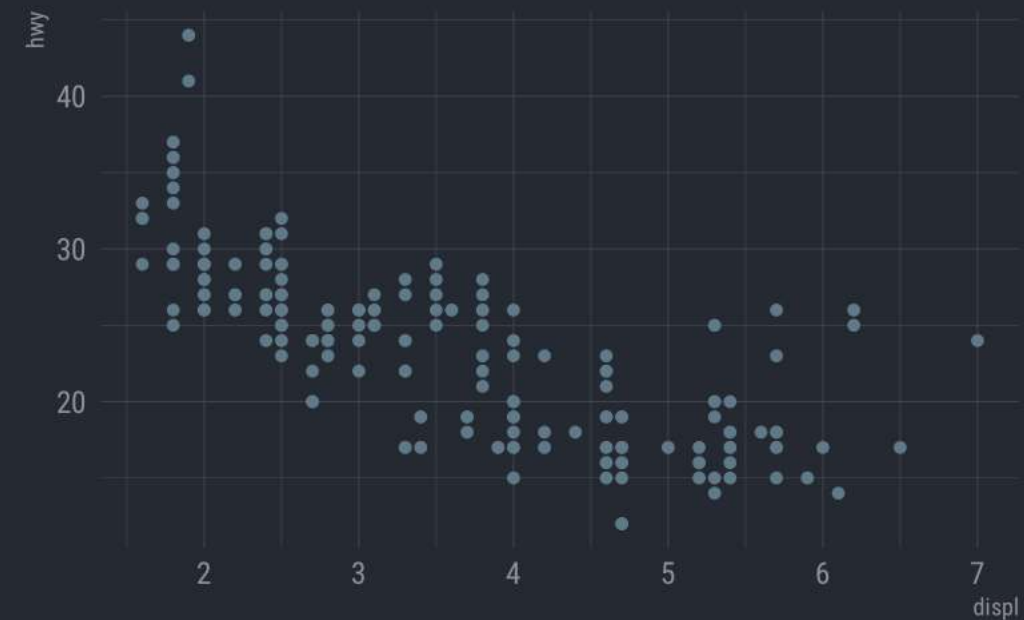
Other themes: [hrbrthemes](#)

```
1 remotes::install_github("hrbrmstr/hrbrthemes")
```

```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point() +  
4   hrbrthemes::theme_ipsum()
```



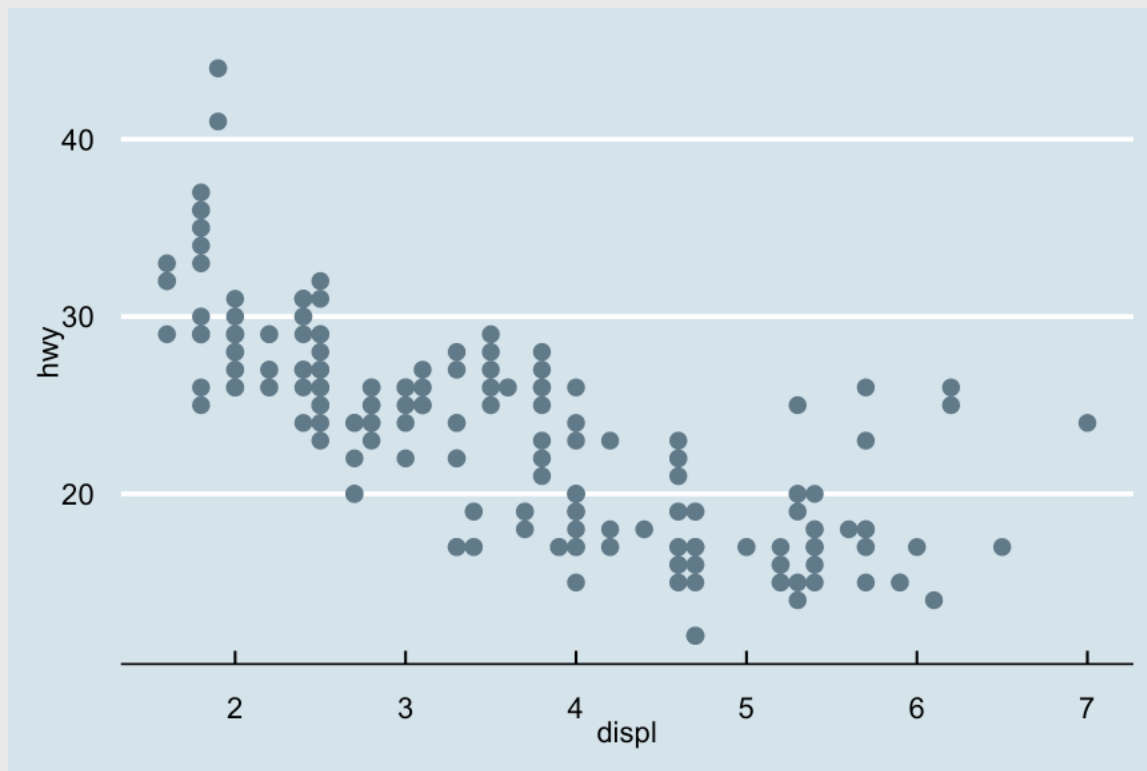
```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point() +  
4   hrbrthemes::theme_ft_rc()
```



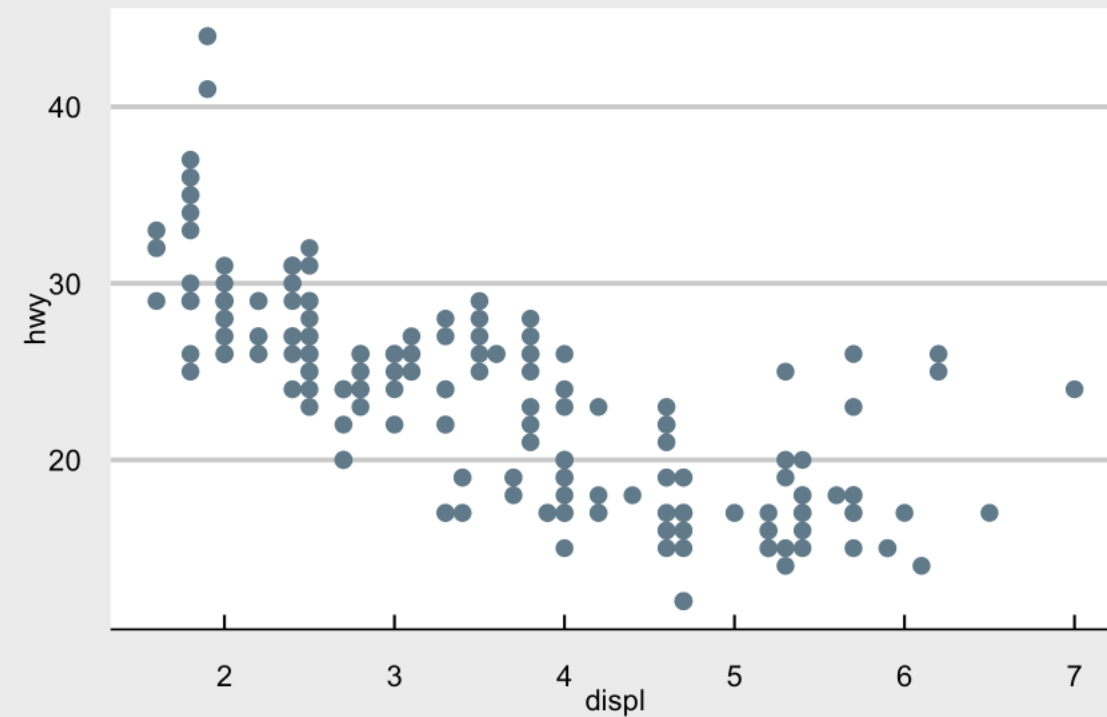
Other themes: `ggthemes`

```
1 install.packages('ggthemes', dependencies = TRUE)
```

```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point() +  
4   ggthemes::theme_economist()
```



```
1 mpg %>%  
2   ggplot(aes(x = displ, y = hwy)) +  
3   geom_point() +  
4   ggthemes::theme_economist_white()
```



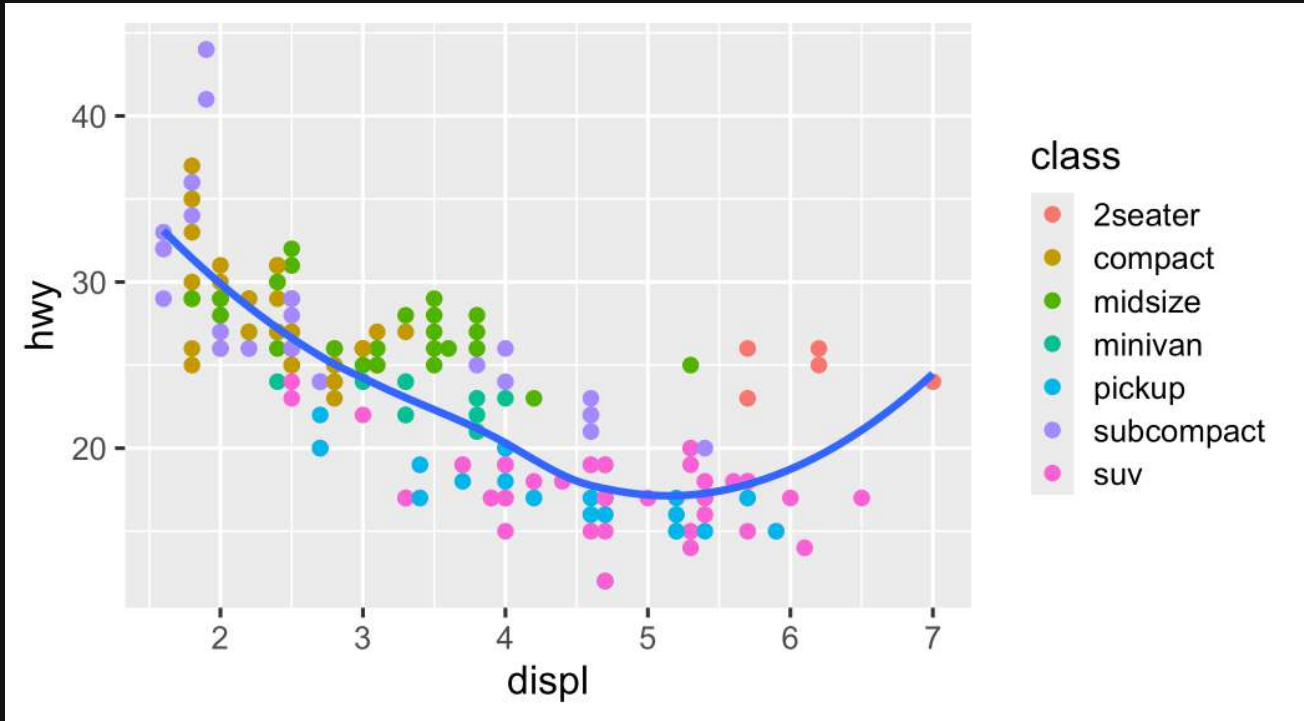
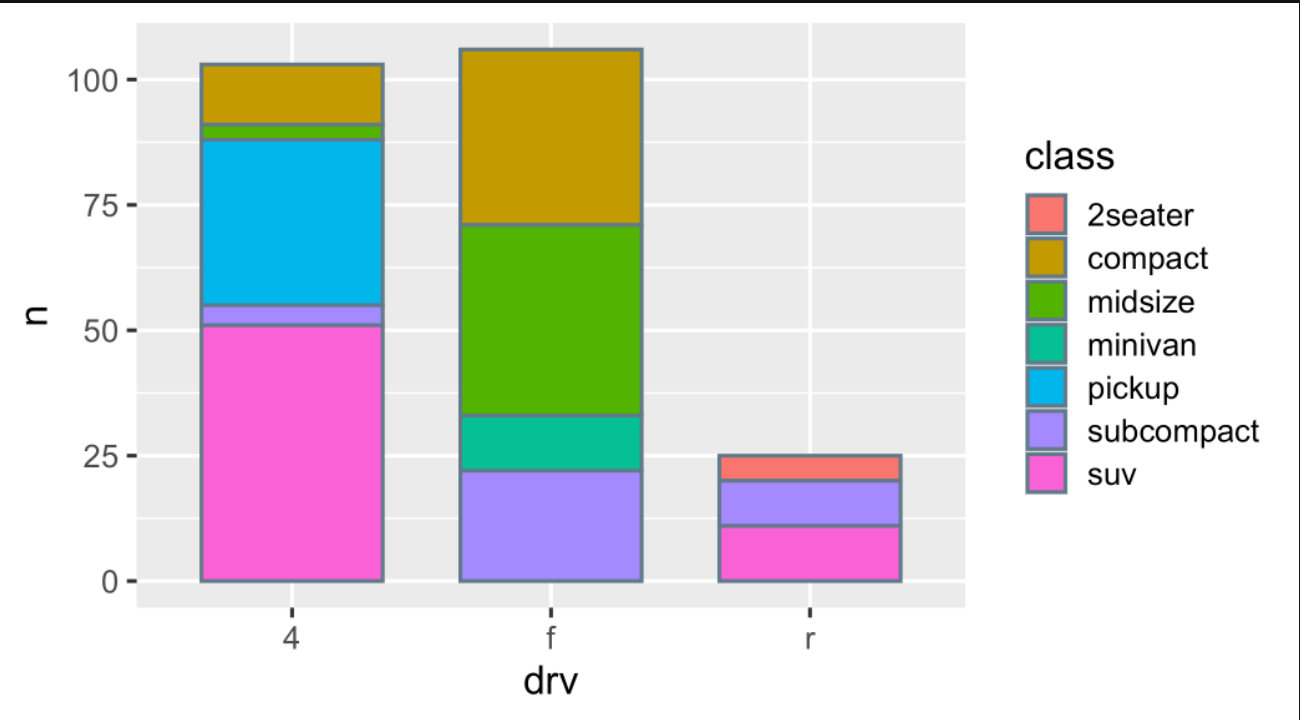
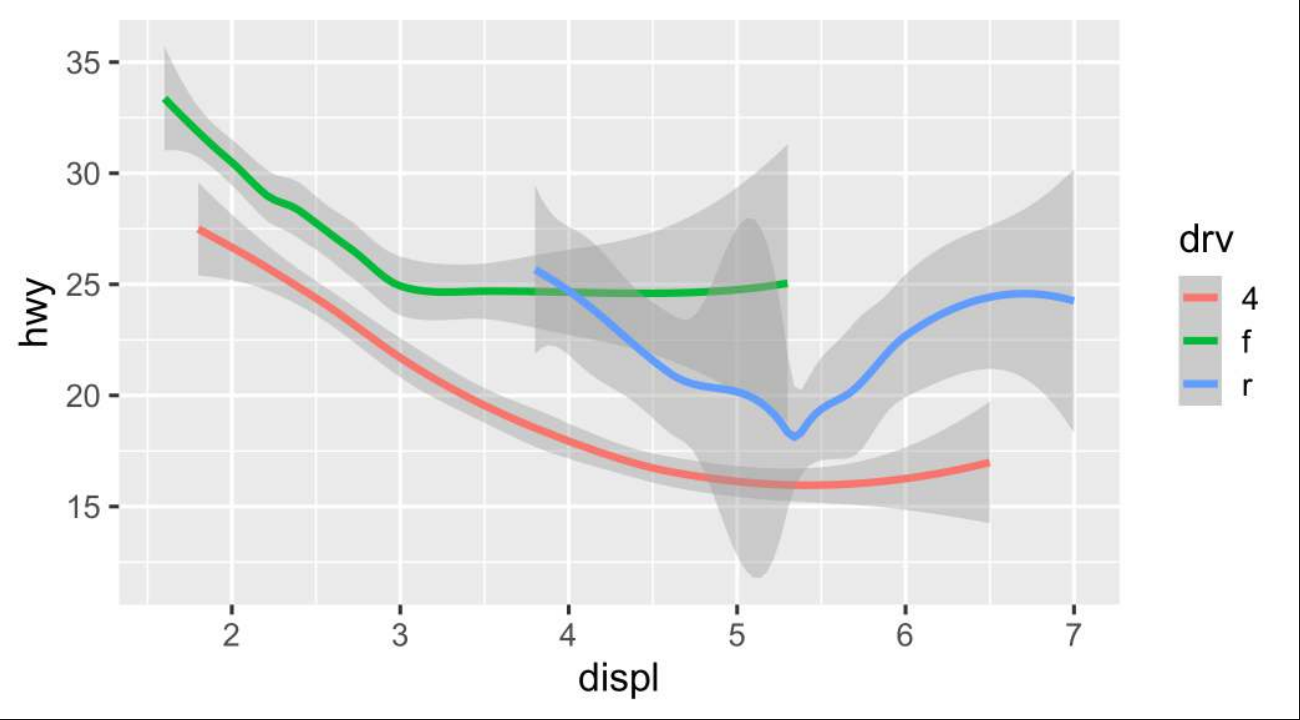
More practice

Open `ggplot2.qmd`

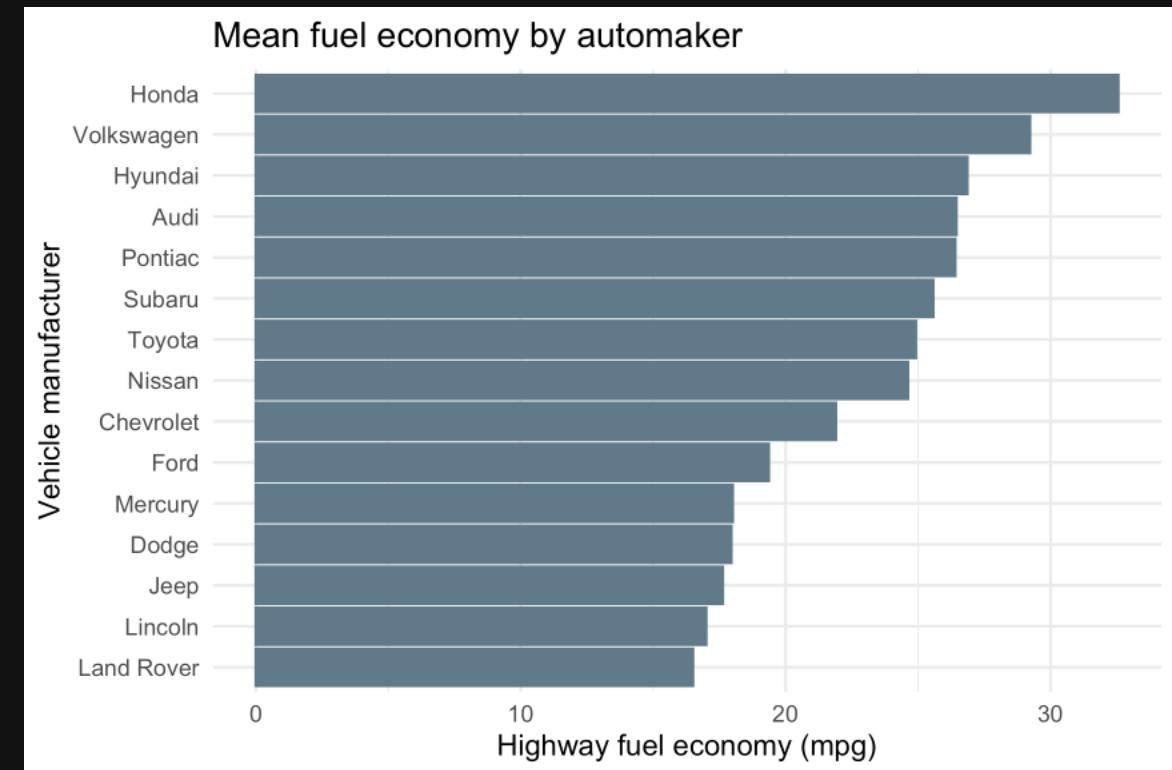
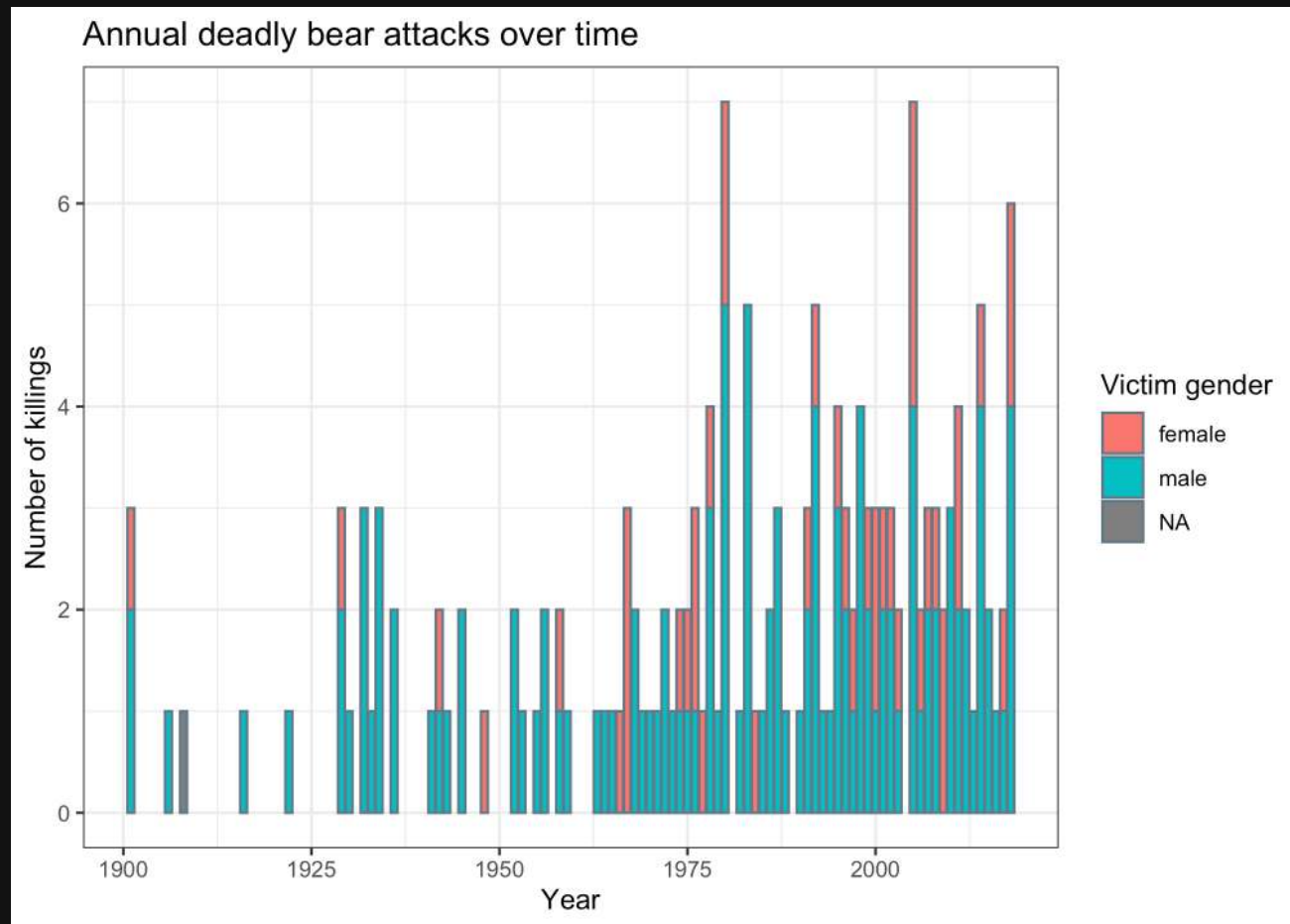
15:00

Your turn

In the "Practice" section of `ggplot2.qmd`
Use the `mpg` data frame and `ggplot` to create these charts



Extra practice



Break

05 : 00

Week 5: *Quarto & Plotting*

1. Intro to Quarto
2. Intro to ggplot2
3. Project attributes & levels
4. Conjoint survey components

Model Relationships Table (example)

		<i>Features</i>			<i>Competitors</i>	
		Range	Units	Demand	Aims Solar Panel	SUAOKI Solar Charger
<i>Product Attributes</i>	Price	\$60 - \$225	USD / kW	-	225	160
	Weight	1 - 3	kg	-	2.6	2.06
	Power Output	100 - 500	W	+	120	60
	Durability	12 - 60	months	+	60	12
	Portability	200 - 2800	cm^3	+	20.6"x11"x 1.2"	11.5"x7.1"x2.9"

Start defining attribute *levels*

Defining attribute levels

Continuous

- **Price:** 1, 2, 3, 4, 5 (\$)
- **Power Output:** 60, 80, 120 (Watts)

Discrete

- **Color:** Red, Blue, Yellow
- **Material:** Plastic, Aluminum, Glass

- Look at competitors
- Search web for values that cover the full set of values available today (and maybe some into the future)

Week 5: *Quarto & Plotting*

1. Intro to Quarto
2. Intro to ggplot2
3. Project attributes & levels
4. Conjoint survey components

3 Parts

- **Part 1:** Intro
- **Part 2:** Conjoint questions
- **Part 3:** Other / demographic questions

3 Parts

- **Part 1:** Intro → screen for target population
- **Part 2:** Conjoint questions → screen for random answers
- **Part 3:** Other / demographic questions

Think of your survey as a *conversation*

- Include “transition” pages (e.g. Great job! Now we’ll ask you about...)

Part 1: Intro

Start with a welcome page

Welcome!

Thank you for participating in this survey. With your responses, we are looking forward to understanding your perspective as a consumer. This research is part of an effort from the George Washington University.

>>

Consent form

This survey is being conducted by students at the George Washington University. We will not be collecting any identifying data such as your name or address. The whole survey will take approximately 10 to 15 minutes to complete. Your participation is voluntary and you may stop the survey at any time.

If you would like to participate, please answer the following questions:

I am age 18 or older

Yes

☐

No

☐

I have read and understand the above information

Yes

☐

No

☐

Eligibility questions: who is your target population?

Filter out respondents here

Are you in the market for a car?

No	☐
Yes, and I plan to buy one in the next few months	☐
Yes, and I plan to buy one in within the next year	☐

>>

Part 2: Conjoint questions

Education information

In this survey, we are going to ask you about your preferences for the following vehicle attributes:

- Price
- Fuel Economy
- Acceleration Time
- Powertrain

But before we ask you any question, let's learn a little bit more about each of these attributes.

Education information

Price is the final price you will pay for the vehicle you are buying
(including all taxes and fees)



>>

POWERTRAIN

The type of engine in the vehicle

Gasoline



Gasoline engine vehicle.

Electric



Electric motor vehicle.
Must be plugged into an
electrical outlet to be
refueled. (6 – 10 hours to
fully charge).

Can be helpful to provide relative comparisons

Weight:

- 1/2 lbs (similar to 1 cup water)
- 8 lbs (similar to 1 gallon of milk)

Conjoint intro

In the next few pages, we will show you options of different vehicles and we'll ask you to select which you would choose.

Let's try a practice question - on the next page, click the round button under the option you most prefer.

A blue rectangular button with rounded corners, containing the white text '>>' in the center.

Practice conjoint (also attention check)

May also filter out respondents here

	Choice 1	Choice 2	Choice 3
Price	20	15	25
Fuel Economy	20	30	15
Acceleration Time	7	6	8
Powertrain			
	☐	☐	☐

Transition to actual conjoint questions

Nice work!
Now let's begin the choice tasks.
You will be asked four choice questions in total

>>

Conjoint questions

May also filter out respondents at the end
(e.g. chose all same answers)

(1 of 4) Choose your preferred option from the choices below:

	Choice 1	Choice 2	Choice 3	None
Price	25	25	15	None of these options
Fuel Economy	30	25	25	
Acceleration Time	8	7	7	
Powertrain				
	☐	☐	☐	☐

>>

Part 3: Other / demographic questions

Transition

Almost done!

We'd like to ask just a few more questions about you.

>>

Critical respondent information

How many vehicles does your household currently own?

>>

Demographic / other questions

What is your age?

What is your total annual household income?

What is your gender?

Which of the following best describes your highest achieved education level?

Finale

We thank you for your time spent taking this survey.
Your response has been recorded.

Project survey plan
(Due in 6 days - next Tuesday)